

Quantum Chaos and Fermionic non-Gaussianity

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How large can fermionic magic be?

Fermionic magic Rényi entropy (MRE)

$$N = 2m \quad (D = 2^m, \quad d = D/2) \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij} \quad (i, j \in [N])$$

$\psi \in \mathcal{H}_+$ even parity sector

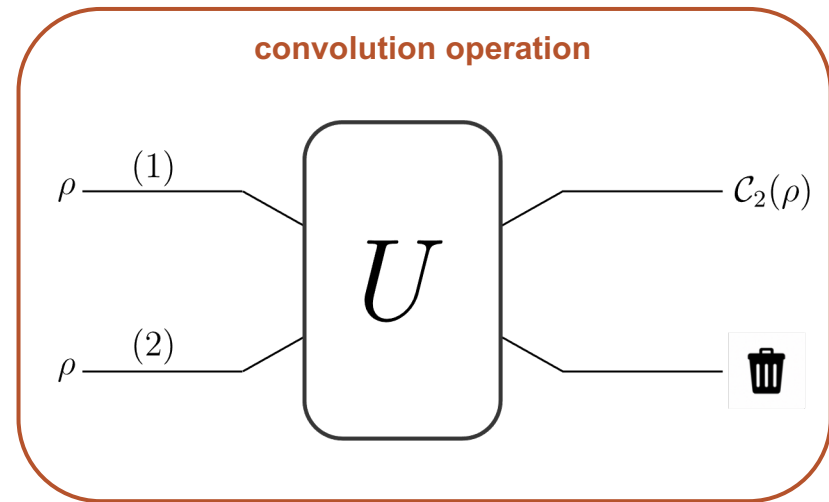
Entanglement created by convolution operation

$$\mathcal{C}_2(\rho) = \text{Tr}_2[U(\rho \otimes \rho)U^\dagger] \quad U = \prod_{i=1}^N \exp\left[\frac{\pi}{8}\gamma_i^{(1)}\gamma_i^{(2)}\right]$$

$$\nu_2 = \text{Tr}[\mathcal{C}_2(\psi)^2], \quad M_2(\psi) = -\ln \nu_2$$

Properties

- **Faithful:** $M_2 \geq 0$; $M_2 = 0$ iff ψ : Gaussian
- **Gaussian invariance:** $M_2(\mathcal{U}_G\psi) = M_2(\psi)$
- **Additive:** $M_2(\psi \otimes \phi) = M_2(\psi) + M_2(\phi)$
- **Monotone under deterministic pure-to-pure Gaussian protocols**



Maximally entangled state

$$|\Psi\rangle = \frac{1}{\sqrt{\chi}} \sum_{j=1}^{\chi} |a_j\rangle |b_j\rangle \quad \text{saturates the upper bound} \quad S_2 = \ln \chi$$

Haar random state: $\overline{S_2} \sim \ln \chi - \ln 2 + o(1)$ ($\dim A = \dim B$)

asymptotically maximal!

Haar random states attain the maximal MRE *density*

Purity calculation using single copy expectation values

$$\nu_2 = \frac{i^m}{2^N} \sum_{c: |A_a| \text{ even}} (-1)^{\text{inv}(c)} \prod_{a=1}^4 \langle \psi | \Gamma_{A_a} | \psi \rangle$$

$$c: [N] \rightarrow \{1, 2, 3, 4\}, \quad A_a = c^{-1}(a), \quad \Gamma_{A_a} = \prod_{i \in A_a} \gamma_i$$

Weingarten formula

$$\mathbb{E}[\rho^{\otimes 4}] = \frac{\sum_{\pi \in S_4} P_\pi}{d(d+1)(d+2)(d+3)}$$

Exact formula for Haar random states

$$\bar{\nu} := \mathbb{E}[\nu_2] = \frac{12D^3 + 16D^2 + 48D + 8 \cdot 3^N + 56}{D(D+2)(D+4)(D+6)} \sim 8 \left(\frac{3}{4} \right)^N \left[1 + \frac{3}{2} \left(\frac{2\sqrt{2}}{3} \right)^N + \mathcal{O}(2^{-N/2}) \right]$$

self-averaging: quenched and annealed averages coincide asymptotically $\frac{\text{Var}(\nu_2)}{\bar{\nu}^2} = 144 \left(\frac{25}{81} \right)^{N/2} + \dots$

Asymptotic scaling of the MRE

$$\mathbb{E}[M_2(\text{Haar})] = N \ln(4/3) - \ln 8 + o(1)$$

achieves maximal MRE density



Theorem: nontrivial upper bound on the MRE

$$M_{\max} := \max_{\psi \in \mathcal{H}_+} M_2(\psi)$$

$$-\ln \bar{\nu} \leq M_{\max} \leq N \ln(4/3) - \ln(16/9)$$

$$\lim_{N \rightarrow \infty} M_{\max}/N = \lim_{N \rightarrow \infty} \mathbb{E}[M_2(\text{Haar})]/N = \ln(4/3)$$

Proof idea

Apply the beam splitter one pair at a time.

- Each step retains at least 3/4 of the purity
- The purity increases by 4/3 at the final step

$$1 \xrightarrow{\times \frac{3}{4}} \frac{3}{4} \xrightarrow{\times \frac{3}{4}} \dots \xrightarrow{\times \frac{3}{4}} \left(\frac{3}{4} \right)^{N-1} \xrightarrow{\times \frac{4}{3}} \left(\frac{3}{4} \right)^{N-2} \leq \nu_2(\psi).$$

Constructing states beyond the Haar mean

Real states

$$N \in 8\mathbb{N}, \quad T := \gamma_1 \gamma_3 \cdots \gamma_{N-1} K, \quad (T^2 = 1)$$

K : complex conjugation

$$T|e_a\rangle = |e_a\rangle, \quad |\psi\rangle = \sum_a x_a |e_a\rangle, \quad (x_a \in \mathbb{R})$$

Covariance matrix of real states is zero

$$T|\psi\rangle = |\psi\rangle \implies C_{ij} = \frac{i}{2} \langle [\gamma_i, \gamma_j] \rangle = 0$$

1 Choose N such that the mean purity of real states is lower than Haar random states

$\bar{\nu}^R$: mean purity on the real unit sphere

$$\frac{\bar{\nu} - \bar{\nu}^R}{\bar{\nu}} = -12 \left(\frac{\sqrt{5}}{3} \right)^N \cos(N \arctan(1/2)) + \mathcal{O}(2^{-N/2})$$

Infinitely many N satisfying

$$\bar{\nu}^R < b_N := \frac{\bar{\nu} + \bar{\nu}^R}{2}, \quad -\ln b_N > \mathbb{E}[M_2(\text{Haar})]$$

2 Replace random variables by integer values

$$|v(x)\rangle = \sum_a x_a |e_a\rangle : \text{unnormalized vector}$$

$$\text{Polynomial of } x: P_N(x) := \|x\|^8 \left[\nu_2 \left(\frac{v(x)}{\|x\|} \right) - b_N \right]$$

$$\mathbb{E}[P_N(x)] = d(d+2)(d+4)(d+6)(\bar{\nu}^R - b_N) < 0$$

choose x from nine integers with probability specified to match Gaussian moments through order eight

3 Choose amplitudes deterministically

1. For all x_1 , average P_N over all remaining coefficients choose the smallest average
2. With x_1 fixed, choose x_2 in the same way
3. Repeat until every coefficient is fixed

The minimum P_N cannot exceed the average

$$P_N(q_1, q_2, \dots, q_d) < 0 \implies \nu_2(\psi) < b_N$$

$$|\psi_N\rangle = \frac{\sum_a q_a |e_a\rangle}{\sqrt{\sum_a q_a^2}}, \quad q_a \in \{-4, -3, \dots, 4\}$$

$$M_2(\psi_N) > \mathbb{E}[M_2(\text{Haar})] \quad \text{for infinitely many } N$$

Same covariance matrix, different MRE—comparison with other measures

Fermionic magic measure via the covariance matrix

Fermionic anti-flatness (FAF): $\mathcal{F}_r = m - \frac{1}{2} \text{Tr}[(C^\top C)^r]$, ($r \geq 1$)

can be generalized to “occupation number entropy”

Tarabunga et al., Computable fermionic non-Gaussianity from the covariance matrix, arXiv:2607.02242

- ✓ Faithful
- ✓ Additive
- ✓ Invariant under Gaussian unitary
- ✓ Monotone for $r = 1$

Identical FAF, extensive difference in the MRE $N = 2m \in 8\mathbb{N}$

Cat state $|\text{cat}_m\rangle := \frac{|0^m\rangle + |1^m\rangle}{\sqrt{2}}$

$$C = 0 \implies \mathcal{F}_r = m \text{ (max)}$$

$$\nu_2 = \frac{1}{8} + \frac{7}{2^{m+1}}, \quad M_2 \rightarrow \ln 8 \quad (N \rightarrow \infty)$$

Constructed real states

$$T|\psi_N\rangle = |\psi_N\rangle \implies C = 0 \implies \mathcal{F}_r = m$$

$$M_2(\psi_N) = N \ln(4/3) + \mathcal{O}(1) \quad (N \rightarrow \infty)$$



extensive difference

Remark: equal MRE but different covariance matrix

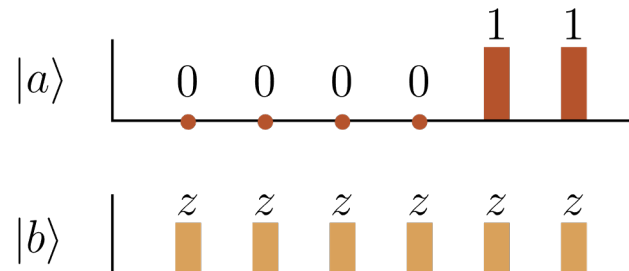
$$|a\rangle = |\text{cat}_4\rangle \otimes |00\rangle, \quad z = \sqrt{\frac{2\sqrt{65}}{5}} - 3,$$

$$|b\rangle = \sqrt{\frac{1+z}{2}} |0^6\rangle + \sqrt{\frac{1-z}{2}} |1^6\rangle$$

$$M_2(a) = M_2(b) = \ln(32/11)$$

$$\mathcal{F}_r(a) = 4 < 6(1 - z^{2r}) = \mathcal{F}_r(b)$$

singular values of the covariance matrix



Fermionic magic generated by SYK dynamics

Sachdev-Ye-Kitaev (SYK) model

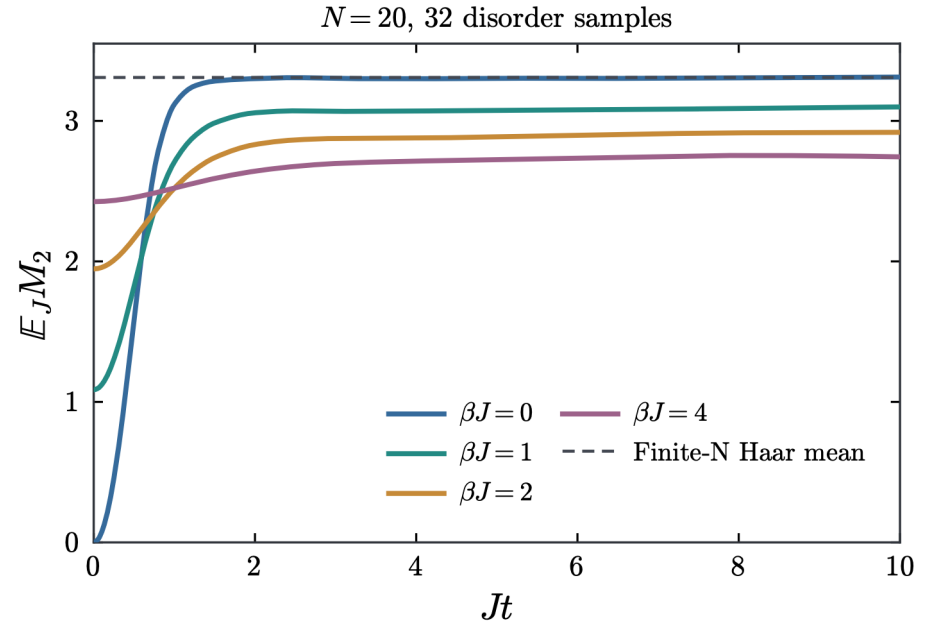
$$H = \sum_{i < j < k < l} J_{ijkl} \gamma_i \gamma_j \gamma_k \gamma_l$$

$$\mathbb{E}_J[J_{ijkl}] = 0, \quad \mathbb{E}_J[J_{ijkl}^2] = 6J^2/N^3$$

Kourkoulou-Maldacena (KM) state

$$|\psi(\beta, t)\rangle = \frac{e^{-wH} |B\rangle}{\sqrt{Z_B(\beta)}}, \quad w = \beta/2 + it$$

$$Z_B(\beta) = \langle B | e^{-\beta H} | B \rangle$$



MRE from four-copy partition functions

partition function of the replicated theory $Z_{\mathcal{R}} = \langle B |^{\otimes 4} e^{-\bar{w} H_{\text{tot}}} \mathcal{R} e^{-w H_{\text{tot}}} | B \rangle^{\otimes 4}$ $H_{\text{tot}} = \sum_{a=1}^4 H^{(a)}$

$$\nu_2(\beta, t) = \frac{Z_{\mathcal{R}}}{Z_B(\beta)^4}, \quad M_2(\beta, t) = -\ln Z_{\mathcal{R}} + 4 \ln Z_B(\beta)$$

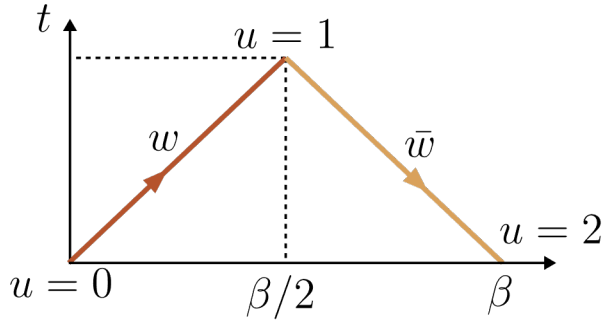
MRE = free energy of the replicated theory with **reflection defect** $\mathcal{R} = (\Pi_3 \Pi_4) (U_{12} U_{34})^\dagger S_{13} (U_{12} U_{34}) (\Pi_3 \Pi_4)$

$$\mathcal{R} \gamma_i^{(a)} \mathcal{R}^\dagger = \sum_{b=1}^4 R_{ab} \gamma_i^{(b)} \quad R = I_4 - \frac{1}{2} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1)^\top \quad \mathcal{R} |B\rangle^{\otimes 4} = |B\rangle^{\otimes 4}$$

Reflection defect intertwines the replicas

Schwinger-Dyson equation at large-N

Reflection on the contour



Parameterization, Rotation of the fields

$$z(u) = \begin{cases} wu, & 0 \leq u \leq 1, \\ w + \bar{w}(u-1), & 1 \leq u \leq 2. \end{cases}$$

$$O(u) = \begin{cases} I_4, & u < 1, \\ R, & u > 1. \end{cases}$$

$$\eta_i(u) = O(u)^\top \chi_i(u)$$

effect of the reflection absorbed
rescaled to $\chi_i = \gamma_i/\sqrt{2}$

KM boundary condition fixes the free propagator F_0

$$F_0(u, v) = \Theta(u - v)I_4, \quad G = \frac{F - F^\top}{2}$$

particle: F hole: $-F^\top$

Collective fields:

$$\hat{G}(u, v) = O(u)G(u, v)O(v)^\top$$

$$G_{ab}(u, v) = \frac{1}{N} \sum_i \eta_i^{(a)}(u) \eta_i^{(b)}(v)$$

Partition function (averaged over random couplings) $\mathbb{E}_J Z_{\mathcal{R}} = \int \mathcal{D}G \mathcal{D}\Sigma e^{-NI[G, \Sigma]}$

$$I[G, \Sigma] = \underbrace{-\frac{1}{2} \ln \det(1 - F_0 \Sigma)}_{\text{integral over fermions } \eta} + \underbrace{\frac{1}{2} \int du dv \sum_{a,b} \Sigma_{ab}(u, v) G_{ab}(u, v)}_{\text{constraint } G_{ab} = \frac{1}{N} \sum_i \eta_i^{(a)} \eta_i^{(b)}} - \underbrace{2J^2 \int du dv \dot{z}(u) \dot{z}(v) \sum_{a,b} \hat{G}_{ab}(u, v)^4}_{\text{from averaging}}$$

$$\mathbb{E}_J e^{\sum_A J_A X_A}, \quad X_A = 4 \sum_a \int_0^2 du \dot{z}(u) \chi_{i_1}^{(a)}(u) \cdots \chi_{i_4}^{(a)}(u)$$

Saddle point (Schwinger-Dyson) equation

$$\delta I / \delta \Sigma = 0 : \quad F = (1 - F_0 \Sigma)^{-1} F_0,$$

$$\delta I / \delta G = 0 : \quad \Sigma(u, v) = 16J^2 \dot{z}(u) \dot{z}(v) O(u)^\top \hat{G}(u, v)^{\circ 3} O(v)$$

$\circ 3$: entrywise cube, $(X^\top)_{ab}(u, v) = X_{ba}(v, u)$

Competing saddles and the MRE transition at large-N

Schwinger-Dyson (SD) equation

$$\begin{aligned} F &= (1 - F_0 \Sigma)^{-1} F_0, \\ \Sigma(u, v) &= 16 J^2 \dot{z}(u) \dot{z}(v) O(u)^\top \hat{G}(u, v) \circ^3 O(v), \\ G &= (F - F^\top)/2, \\ \hat{G} &= O(u) G(u, v) O(v)^\top. \end{aligned}$$



Ansatz for the solution

$$\begin{aligned} G(u, v) &= g(u, v) I_4 + h(u, v) A, \\ g(u, v) &= -g(v, u), \quad h(u, v) = h(v, u), \\ A &= \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 1 \\ -1 & 1 & -1 & 0 \end{pmatrix}, \quad A^2 = -3I_4 \end{aligned}$$

The SD equations preserve this form: $G \rightarrow \Sigma \rightarrow F \rightarrow G$
Determine the two unknown functions $g(u, v)$ and $h(u, v)$

1 Find solutions to the SD equation at (β, t)

$$h = 0$$

keep $h = 0$, solve for $g(u, v)$

S_4 : symmetric under all permutations

$$h \neq 0$$

use nonzero initial h , solve for $g(u, v)$ and $h(u, v)$

C_3 : symmetric under cyclic permutations of 2,3,4

Discretize u, v into $L \times L$ grid and iterate the update (with mixing) until the SD equation converge.

2 Compare actions to identify the dominant saddle

$$\mathbb{E}_J Z_{\mathcal{R}} \sim \sum_{\text{saddles}} e^{-N I[G, \Sigma]}$$

MRE: free energy of the replicated theory $M_2 = -\ln Z_{\mathcal{R}} + 4 \ln Z_B$

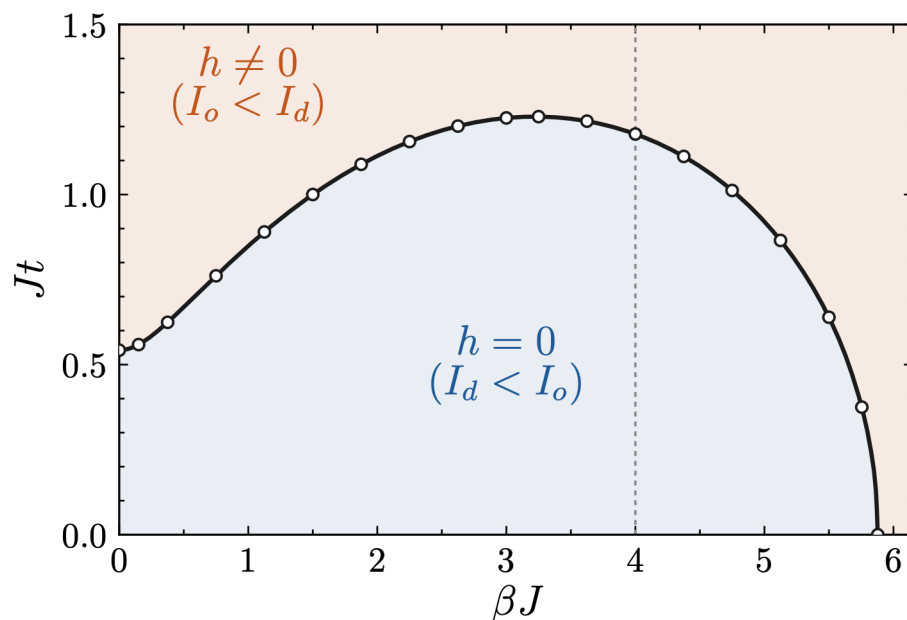
$$\frac{\mathbb{E}_J M_2}{N} \simeq \min \operatorname{Re} I[G, \Sigma] + \frac{4}{N} \mathbb{E}_J \ln Z_B$$

Crossing of the dominant saddles
produce a kink in MRE density

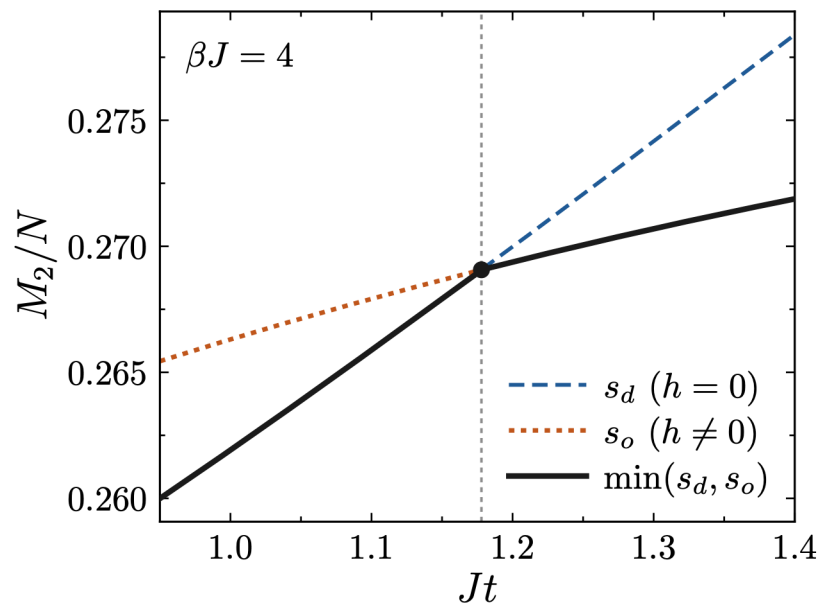
Predicted transition in the MRE

$$s_{d/o} = I_{d/o} - 4I_B, \quad (d : h = 0, \quad o : h \neq 0) \quad I_B \sim -\frac{1}{N} \ln \mathbb{E}_J Z_B \quad M_2/N \rightarrow \min\{s_d, s_o\}$$

$(\beta J, Jt)$ phase diagram



Kink at the transition point



Transition between two saddles

$$S_4 \rightarrow C_3, \quad (h = 0 \rightarrow h \neq 0)$$

Correlations between the replicas appear
MRE density stays continuous

Numerical checks

- Crossing converges under grid refinement up to $L = 2048$
 - 1,152 initial guesses, no converged solution below both candidate actions
- The two saddles dominate at large- N

Summary and Outlook

1 | Haar random states attain the maximal MRE density

$$\mathbb{E}[M_2(\text{Haar})] = N \ln(4/3) - \ln 8 + o(1),$$

$$\lim_{N \rightarrow \infty} \frac{M_{\max}}{N} = \ln \frac{4}{3}$$

An explicit family of real states exceed the Haar mean

$$M_2(\psi_N) > \mathbb{E}[M_2(\text{Haar})]$$

2 | Equal covariance matrix, extensively different MRE

$$\mathcal{F}_r(\text{cat}_{N/2}) = \mathcal{F}_r(\psi_N) \quad [\because C(\text{cat}_{N/2}) = C(\psi_N)]$$

$$M_2(\psi_N) - M_2(\text{cat}_{N/2}) = N \ln(4/3) + O(1)$$

The MRE detects higher order correlations

3 | Transition in the MRE of SYK model dynamics

$$H = \sum_{i < j < k < l} J_{ijkl} \gamma_i \gamma_j \gamma_k \gamma_l \quad |\psi(\beta, t)\rangle = \frac{e^{-(\beta/2 + it)H} |B\rangle}{\sqrt{Z_B(\beta)}}$$

$$M_2/N \rightarrow \min\{s_d, s_o\} \quad d : h = 0 (S_4), \quad o : h \neq 0 (C_3)$$

MRE density is continuous with a kink; jump in the slope

Outlook

Strengthen the large-N prediction

Test saddle dominance and the assumptions about disorder averaging

Test the prediction at finite N

Compute the MRE for larger systems and test whether the slope changes more sharply as N increases

Explore a gravitational interpretation

Do the competing MRE saddles have a geometric interpretation in the gravitational dual?